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K.V.S. Matriculation Hr. Sec. School, Thoothukudi.

Half Yearly Exam

PHYSICS

Time: 3.00 Hours Marks: 70 Std :XII Name:

I. Choose the correct answer : **[15 × 1 = 15]**

1) Two metallic spheres of radii 1 cm and 3 cm are given charges of $-1 \times 10^{-2}\ C$ and $5 \times 10^{-2}\ C$ respectively. If these are connected by a conducting wire, the final charge on the bigger sphere is
a) $3 \times 10^{-2}\ C$ b) $4 \times 10^{-2}\ C$ c) $1 \times 10^{-2}\ C$ d) $2 \times 10^{-2}\ C$

2) A parallel plate capacitor has square plates of side 5 cm and separated by a distance of 1 mm. Calculate the capacitance of this capacitor.
a) $22.12 \times 10^{-32} F$ b) $22.12 \times 10^{+32} F$ c) $22.12 \times 10^{-12} F$ d) $22.12 \times 10^{+12} F$

3) A potential difference across 24 Ω resistor is 12 V. What is the current through the resistor?
a) $0.5\ A$ b) $1\ A$ c) $1.5\ A$ d) $2\ A$

4) In India electricity is supplied for domestic use at 220 V. It is supplied at 110 V in USA. If the resistance of a 60 W bulb for use in India is R, the resistance of a 60 W bulb for use in USA will be
a) R b) $2R$ c) $R/4$ d) $R/2$

5) The magnitude of magnetic moment of an electron in a circular orbit of radius r with a speed V is equal to
a) $\frac{eVr}{2}$ b) eVr c) $\frac{er}{2V}$ d) $\frac{2V}{er}$

6) A circular coil of radius 5 cm and 50 turns carries a current of 3 ampere. The magnet dipole moment of the coil is nearly
a) $1.0\ Am^2$ b) $1.2\ Am^2$ c) $0.5\ Am^2$ d) $0.8\ Am^2$

7) In a transformer, the number of turns in the primary and the secondary coil are 300 and 1800 respectively. If the current in primary is 6 A, then that in the secondary coil is
a) 2 A b) 18 A c) 12 A d) 1 A

8) $\frac{20}{\pi^2}\ H$ inductor is connected to a capacitor of capacitance C. The value of C in order to impart maximum power at 50 Hz is
a) $50\ \mu F$ b) $0.5\ \mu F$ c) $500\ \mu F$ d) $5\ \mu F$

9) If V_g, V_x, V_m are speeds of gamma rays, X-rays and microwaves respectively in vacuum, then
a) $V_g < V_x < V_m$ b) $V_g > V_x > V_m$ c) $V_g > V_x < V_m$ d) $V_g = V_x = V_m$

10) The dimension of $\frac{1}{\mu_0 \varepsilon_0}$ is
a) $[LT^{-1}]$ b) $[L^2T^{-2}]$ c) $[L^{-1}T]$ d) $[L^{-2}T^{-2}]$

11) A ray of light gets refracted into the air medium from crown glass of refractive index 1.541. If angle of incidence is equal to the critical angle 40.5^0, then the angle of refraction will be
a) equal to the critical angle b) lesser than the critical angle
c) equal to 90^0 d) greater than critical angle

12) Pure water has refractive index 1.33. What is the speed of light through it?
a) $2.26 \times 10^8\ ms^{-1}$ b) $2.26 \times 10^{-8}\ ms^{-1}$ c) $12.26 \times 10^8\ ms^{-1}$ d) $12.26 \times 10^{-8}\ ms^{-1}$

13) For a Astigmatism eye, the defect is cured by using a
a) convex lens b) concave lens c) cylindrical lens d) plane glass

14) Two coherent monochromatic light beams of intensities I and 4I are superposed. The maximum and minimum possible intensities in the resulting beam are
a) 5I and I b) 5I and 3I c) 9I and I d) 9I and 3I

15) The threshold wavelength for a metal surface whose photoelectric work function is 3.313 eV is
a) 4125 Å b) 3750 Å c) 6000 Å d) 2062.5 Å

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II) Answer any Six questions. Question no. 24 is compulsory: **[6× 2 = 12]**

16) Define 'electric flux'
17) Define electrical resistivity.
18) State Ampere's circuital law.
19) What are step-up and step-down transformers?
20) What is displacement current?
21) What is principle of reversibility?
22) What is astigmatism? What is its remedy?
23) Define stopping potential.
24) In an electric field of magnitude 570 NC^{-1}, is applied in the copper wire, find the acceleration experienced by the electron.

III) Answer any Six questions. Question no. 33 is compulsory: **[6× 3 = 18]**

25) Explain in detail Coulomb's law and its various aspects.
26) Obtain the condition for bridge balance in Wheatstone's bridge.
27) Give an account of magnetic Lorentz force.
28) Mention the various energy losses in a transformer.
29) Explain the types of absorption spectrum.
33) What is the focal length of the combination if the lenses of focal lengths -70 cm and 150 cm are in contact? What is the power of the combination?
31) Discuss about pile of plates.
32) Derive an expression for de Broglie wavelength of electrons.
33) A radiation of wavelength 300 nm is incident on a silver surface. Will photoelectrons be observed? [work function of silver = 4.7 eV]

IV) Answer all the questions in detail: **[5 × 5 = 25]**

34) a) Explain in detail the construction and working of a Van de Graaff generator.

(OR)

b) Describe the microscope model of current and obtain general form of Ohm's law.

35) a) Calculate the magnetic field inside and outside of the long solenoid using Ampere's Circuital law.

(OR)

b) Derive an expression for phase angle between the applied voltage and current in a series RLC circuit

36) a) Write down the properties of electromagnetic waves.

(OR)

b) Obtain the lens maker's formula and mention its signification.

37) a) Obtain the equation for resolving power of optical instruments.

(OR)

b) Briefly explain the principle and working of electron microscope.

38) a) Discuss the conversion of galvanometer into an ammeter and also a voltmeter.

(OR)

b) Prove that the total energy is conserved during LC oscillations.

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